

Final Model Comparison Report

This report summarizes the performance of all regression models evaluated during the project using MAE, MSE, and R² metrics.

Model Performance Table (Validation Set):

Model	MAE (Val)	MSE (Val)	R ² (Val)
Linear Regression	325.64	1552248.17	0.95
Decision Tree	162.01	1136255.62	0.96
Random Forest	160.06	1004541.83	0.97
Gradient Boosting	338.45	1182456.38	0.96
KNN	198.64	1965695.98	0.93
Voting Ensemble	249.67	1008388.69	0.97
Average Ensemble	223.25	1099540.66	0.96
Bayesian Ridge	283.62	1250871.33	0.96

Model Performance Summary:

The best performing model was **Random Forest**, which achieved the highest R² score on the validation set. This indicates that it captured variance in the target variable more effectively than the other models.

The worst performing model was **KNN**, which produced the lowest R² score. This suggests it struggled to generalize or fit the underlying patterns in the data.

Overall, ensemble methods tended to perform strongly, especially models that combine predictions from multiple learners.